

THE IMPORTANCE OF FIBER AS A SOURCE OF ENERGY DURING THE TRANSITION PERIOD

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INTRODUCTION

Carbohydrates comprise the majority of diets fed to transition cows and are the most important source of energy during this timeframe as in other parts of the lactation cycle. Much of the research conducted in carbohydrate nutrition of transition cows has focused on manipulating energy levels during the prepartum period by varying neutral detergent fiber (NDF) and starch concentrations in prepartum diets (e.g., Janovick and Drackley, 2010; Mann et al., 2015) or varying starch and digestible NDF levels during the postpartum period by varying concentrations of starch and nonforage fiber sources (NFFS) in the ration (e.g., Dann and Nelson, 2011; McCarthy et al., 2015). Comparatively little work has focused directly on the impact of fiber digestibility in diets fed during the prepartum and postpartum periods. In this paper we will summarize our current knowledge in this area and suggest future directions for research to further support additional recommendations for feeding transition cows.

RESEARCH REPLACING FORAGE NDF WITH NONFORAGE NDF IN DIETS FED DURING THE TRANSITION PERIOD

Several studies have been conducted that have focused on replacing forage with nonforage fiber sources in transition diets. Holcomb et al. (2001) fed multiparous cows high (70% forage) vs. low (28% forage) forage diets at two levels of intake for the last 28 d prepartum and fed a common lactation diet postpartum. To formulate the low forage diets, bermudagrass hay and corn silage were replaced by mostly soybean hulls and hominy feed. Concentrations of NDF (39.2 vs. 44.2%) and starch (25.1 vs. 22.0%) were somewhat different between the low forage and high forage diets. Cows fed the low forage diet for ad libitum intake had higher prepartum dry matter intake (DMI; 14.1 vs. 10.7 kg/d), lower overall plasma concentrations of nonesterified fatty acids (NEFA), and higher overall plasma concentrations of insulin; however, postpartum DMI (20.5 vs. 21.2 kg/d) was not different for cows fed low vs. high forage diets, respectively, during the prepartum period and milk yield was numerically higher (35.8 vs. 29.9 kg/d) for cows fed the high forage diet prepartum.

Dann et al. (2007) partially replaced oat hay in corn-silage based prepartum rations with beet pulp for the last 21 d before expected calving fed to multiparous and primiparous cows. Neither prepartum nor postpartum DMI, prepartum and postpartum serum NEFA, or milk yield were affected by treatment.

Chung et al. (2008) fed multiparous cows either a conventional, high forage (70% of DM) ration or a lower forage (42% of DM) ration in which corn silage and grass hay were replaced with cottonseed hulls, soybean hulls, and alfalfa hay for the entire dry period. Feeding the ration higher in NFFS resulted in substantially higher prepartum DMI (15.7 vs 11.9 kg/d) along with higher plasma insulin and lower plasma NEFA concentrations; however, postpartum DMI (21.8 vs. 21.3 kg/d) and milk yield (45.2 vs. 43.4 kg/d) were not different for cows fed the low vs. high forage diets, respectively. Plasma NEFA were not different postpartum for cows fed the different prepartum treatments; however, interestingly, plasma B-hydroxybutyrate (BHBA) concentrations tended to be higher postpartum for cows fed the low forage diet prepartum.

Overall, results suggest that replacing forage NDF with NDF from NFFS sources during the prepartum period is likely to increase DMI during the prepartum period; however, these effects on prepartum DMI do not seem to translate into effects on postpartum DMI and milk yield. The increased prepartum DMI likely does contribute to better metabolic status for cows fed NFFS based upon higher circulating insulin and lower NEFA concentrations.

EFFECTS OF VARYING FORAGE DIGESTIBILITY DURING THE TRANSITION PERIOD

In contrast to results reported above in which replacement of forage NDF with NFFS during the prepartum period resulted in modest effects on postpartum outcomes, improved forage NDF digestibility in diets fed to transition cows generally does result in better postpartum performance and metabolic status.

Stone et al. (2012) replaced conventional corn silage with BMR corn silage in diets fed during the last 3 wk before calving through wk 3 to 4 postpartum, at which time all cows were fed the postpartum diet containing conventional corn silage through 15 wk postpartum. Diets were formulated to be the same in chemical composition and vary only in NDF digestibility. Cows fed BMR corn silage had higher DMI during both the prepartum (14.3 vs. 13.2 kg/d) and 3-wk postpartum (20.1 vs. 18.1 kg/d) periods than cows fed conventional corn silage. Milk yield was increased during the first 3 wk postpartum (37.5 vs. 34.3 kg/d) for cows fed BMR corn silage and there was a carryover response from wk 4 to 15 postpartum in that cows previously fed BMR corn silage tended to make more milk than cows previously fed conventional corn silage (37.3 vs. 35.3 kg/d). Metabolic status as reflected by plasma concentrations of NEFA and BHBA were similar across the two treatments.

Recently, LaCount et al. (2017) conducted a study similar to the one immediately above in that they replaced conventional corn silage with BMR corn silage in rations fed from d 21 before expected calving through 42 d postpartum; furthermore, cows were also randomized to control vs. monensin (0 mg/d vs. target 330 mg/d prepartum and 500 mg/d postpartum) treatments in a 2 x 2 factorial arrangement. In this study, cows fed BMR corn silage had higher prepartum DMI (14.7 vs. 14.0 kg/d) but similar postpartum DMI (23.3 vs. 23.0 kg/d) than cows fed conventional corn silage (Table 1). However, cows fed

BMR corn silage had higher milk yields during the first 42 days in milk (48.3 vs. 45.8 kg/d; Table 1) and had lower plasma concentrations of NEFA during both the prepartum and postpartum periods and lower BHBA during the postpartum period (Table 2).

Table 1. Main effect means for prepartum and postpartum intake and milk yield for cows fed conventional (CON) vs. brown midrib (BMR) during the transition period and 0 (NO) vs. 330 mg/d of monensin prepartum and 500 mg/d of monensin postpartum (MON). From LaCount et al., 2017.

Item	Corn Silage			Monensin			P-Value		
	CON	BMR	SEM	NO	MON	SEM	Corn	MON	C×M×T ¹
Prepartum									
Intake, kg/d	14.0	14.7	0.21	14.5	13.9	0.21	0.03	<0.01	0.85
Postpartum									
Intake, kg/d	23.0	23.3	0.39	23.3	23.0	0.39	0.57	0.55	0.44
Yield, kg/d	45.8	48.3	0.86	46.7	47.4	0.86	0.05	0.61	0.02

¹Interaction of Corn silage × Monensin × Time, no other interactions for these variables were significant.

Table 2. Prepartum and postpartum NEFA and BHBA presented as geometric means with back transformed 95% confidence limits for cows fed conventional (CON) vs. brown midrib (BMR) during the transition period and 0 (NO) vs. 330 mg/d of monensin prepartum and 500 mg/d of monensin postpartum (MON). From LaCount et al., 2017.

Item	Corn		Monensin		P-Value		
	CON	BMR	NO	MON	Corn	MON	C×T ¹
Prepartum							
NEFA, μEq/L	110.6 101.0-121.3	94.8 86.2-104.2	99.3 90.4-109.1	105.6 96.3-115.8	0.02	0.35	0.05
BHBA, mmol/L	0.72 0.68-0.77	0.70 0.66-0.75	0.75 0.70-0.79	0.68 0.64-0.72	0.62	0.04	0.53
Postpartum							
NEFA, μEq/L	458.9 415.7-506.7	369.1 334.2-407.7	440.0 398.4-486.0	385.0 348.7-425.0	<0.01	0.06	0.48
BHBA, mmol/L	1.22 1.10-1.35	1.00 0.90-1.11	1.21 1.10-1.34	1.00 0.91-1.11	<0.01	0.01	<0.01

¹Interaction of Corn × time, no other interactions for these variables were significant.

Evidence from these two trials demonstrates that increased forage fiber digestibility in the periparturient period can positively affect both performance and metabolism throughout the transition period.

TRANSITION COW PERFORMANCE AND METABOLISM AS AFFECTED BY NDF POOLS

During the past several years, the analytical capacity to determine different fractions of NDF (both digestible NDF and undigested NDF at varying timepoints (e.g., uNDF₂₄₀) has become common in commercial forage laboratories. However, little is known specifically about optimum fractions of digestible NDF and uNDF₂₄₀ for the dairy cow during the immediate postcalving period.

Several years ago, we developed a case study within the context of an experiment designed to assess responses in higher vs. lower starch diets fed during the postcalving period (detailed in McCarthy et al., 2015). Cows originally enrolled in the experiment developed significant metabolic issues postcalving (primarily displaced abomasum and clinical ketosis). Upon investigation, we discovered that the NDF content of both the BMR corn silage and legume silage fed had decreased significantly since the samples used to formulate diets were collected. We modified the rations to decrease the inclusion rate of BMR corn silage and increase wheat straw (Table 3).

Table 3. Ingredient and chemical composition of diets before and after postpartum ration changes (adapted from McCarthy et al., 2015).

Item	Postpartum diet ¹			
	HSLF	LSLF	HS HF	LS HF
Ingredient, % of DM				
BMR corn silage	46.1	46.1	38.5	38.5
Wheat straw	3.8	3.8	11.5	11.5
Legume silage	9.6	9.6	9.6	9.6
Higher starch grain mix	40.5	...	40.4	...
Lower starch grain mix	...	40.5	...	40.4
Chemical ²				
NDF, %	26.4	31.5	34.3	36.9
Starch, %	28.3	22.0	26.2	21.5
30-h dNDF, % of NDF	56.0	59.8	51.4	51.7
30-h dNDF, % of DM	15.9	20.4	17.5	18.5
uNDF ₂₄₀ , % of DM	7.7	8.9	10.5	10.9

¹ HSLF = high starch, low fiber (pre-change); LSLF = low starch, low fiber (pre-change); HSHF = high starch, high fiber (post-change); LSHF = low starch, high fiber (post-change).

² Determined using wet chemistry techniques for all variables (Cumberland Valley Analytical Services, Hagerstown, MD).

Performance and metabolic responses for cows fed higher and lower starch diets before and after the changes to both rations are presented in Table 4. The statistics and results should be interpreted with caution as the comparisons for the fiber main effect were not contemporaneous. However, an interaction of starch and fiber suggested that cows fed the combination of higher starch and high fiber had the highest

DMI, the combination of higher starch and low fiber had the lowest DMI, and DMI for cows fed the other two treatments was intermediate. Increased plasma concentrations of NEFA and BHBA for cows fed the combination of higher starch and low fiber were consistent with the lower DMI and poorer energy status of this group. Furthermore, concentrations of haptoglobin were increased for cows fed low fiber, suggesting greater systemic inflammation. Intakes of uNDF₂₄₀ were distinctly different for cows fed the low- and high fiber and both the uNDF₂₄₀ and 30-h dNDF values were markedly different between the low- and high fiber diets (Table 3).

Table 4. Postpartum DMI, uNDF₂₄₀ intakes, milk yield, and Ingredient and chemical composition of diets before and after postpartum ration changes (adapted from McCarthy et al., 2015).

Item	Postpartum diet ¹				Effect ²
	HSLF	LSLF	HS HF	LS HF	
DMI, kg/d	17.8	19.9	21.4	20.1	F, S*F
uNDF ₂₄₀ intake, kg/d	1.52	1.67	2.25	2.15	F, S*F
uNDF ₂₄₀ intake, % of BW	0.26	0.28	0.37	0.36	F, S*F
Milk, kg/d	34.5	40.8	39.2	37.9	S*F
NEFA, uEq/L	646	528	406	493	F, S*F
BHBA, mg/dL	12.3	8.9	9.3	10.7	S*F
Haptoglobin, g/L	1.43	1.71	1.03	0.87	F
Displaced abomasum	4/7	2/10	0/38	0/39	
Clinical ketosis	4/7	1/10	4/38	6/39	

¹ HSLF = high starch, low fiber (pre-change); LSLF = low starch, low fiber (pre-change); HSHF = high starch, high fiber (post-change); LSHF = low starch, high fiber (post-change).

² F = effect of fiber (pre-change vs. post-change); S*F = interaction of starch and fiber.

These results led us to want to explore the relationships of NDF pools in postpartum diets in a more controlled manner, so LaCount et al. (2017) formulated diets to be either low fiber or high fiber through replacement of conventional corn silage with straw (Table 5). Respective treatment diets were fed through 28 DIM, after which all cows were fed the low fiber diet. In this case, cows fed the high fiber diet had lower postpartum DMI along with lower milk yield during the immediate postpartum period (Table 6). Furthermore, cows fed high fiber had higher plasma NEFA, higher BHBA, lower glucose, and a greater degree of negative energy balance while consuming the high fiber diet. After 28 DIM once cows assigned the high fiber diet were switched to the lower fiber diet, the cows were able to quickly meet the performance and metabolic status of cows fed low fiber throughout the study. Clearly, in this experiment we exceeded the uNDF₂₄₀ needs of the cows fed the high fiber diet such that we negatively influenced DMI and metabolism. Two important differences between this experiment and the previous case study were that we utilized conventional corn silage as the base forage in the controlled study and that retrospective analysis of the alfalfa hay that we utilized revealed that this particular hay had a uNDF₂₄₀ value of about 50% of the NDF. This increased the uNDF₂₄₀ value of both postpartum diets in excess of our targets.

Table 5. Ingredients and nutrient profile of rations (mean $\pm$ SD), obtained through wet chemistry analysis and in vitro fermentation (LaCount et al., 2017).

Item	Diet		
	Prepartum	Low Fiber (LF)	High Fiber (HF)
Ingredients, % of ration DM			
Conventional corn silage	45.21	42.31	38.46
Alfalfa hay	-	10.58	10.58
Straw	20.84	1.15	8.65
Corn meal	2.43	17.64	20.15
Soybean meal	-	6.03	4.73
Wheat middlings	-	4.82	1.58
Amino Plus	5.9	4.34	5.31
Canola meal	3.47	1.61	3.88
Corn gluten feed	1.74	1.61	0.47
Blood meal	2.43	0.95	1.09
Soybean hulls	6.95	2.41	-
Citrus pulp	4.52	-	0.79
Energy Booster	-	1.29	1.58
Rumensin, mg/d ¹	439	365	334
Other	6.4	2.3	2.3
Analyses, % of ration DM			
aNDFom	43.1 $\pm$ 0.3	32.8 $\pm$ 1.4	35.3 $\pm$ 2.3
ADF	29.0 $\pm$ 0.5	21.3 $\pm$ 1.1	22.9 $\pm$ 2.1
Starch	15.6 $\pm$ 0.3	24.8 $\pm$ 1.7	24.6 $\pm$ 2.3
Sugar	3.5 $\pm$ 0.4	5.0 $\pm$ 0.7	3.9 $\pm$ 0.1
Fat	2.3 $\pm$ 0.2	3.3 $\pm$ 0.2	3.2 $\pm$ 0.2
uNDF ₂₄₀	12.8 $\pm$ 0.5	9.5 $\pm$ 0.4	12.2 $\pm$ 1.6
peNDF	33.3	21.6	23.2
MP, g/kg DM ¹	89.0	112.1	108.0

¹ Formulated value given by Cornell Net Carbohydrate and Protein System v. 6.55 using actual mean intakes

Table 6. The effect of low fiber and high fiber diets in the early postpartum period on intake, milk yield, and rumination for 1 to 6 wk postpartum (LaCount et al., 2017).

Item	LF	HF	SEM	P-Value	
				Trt	Trt×Time
Prepartum DMI, kg/d		15.5		-	-
Postpartum DMI, kg/d	23.6	22.2	0.3	0.002	<0.001
uNDF intake, %BW ¹	0.29	0.34	0.01	<0.001	0.047
Milk yield, kg/d	46.2	44.7	1.0	0.26	0.001
Fat, %	3.72	3.87	0.85	0.20	0.14
Protein, %	3.09	3.02	0.05	0.30	0.56
Total solids, %	12.57	12.64	0.13	0.70	0.33
ECM, kg/d	48.2	47.3	1.1	0.55	0.12
Rumination, min/d	544	543	8	0.90	0.14

¹Calculated only for weeks where treatment diets differ (1 to 4 wk)

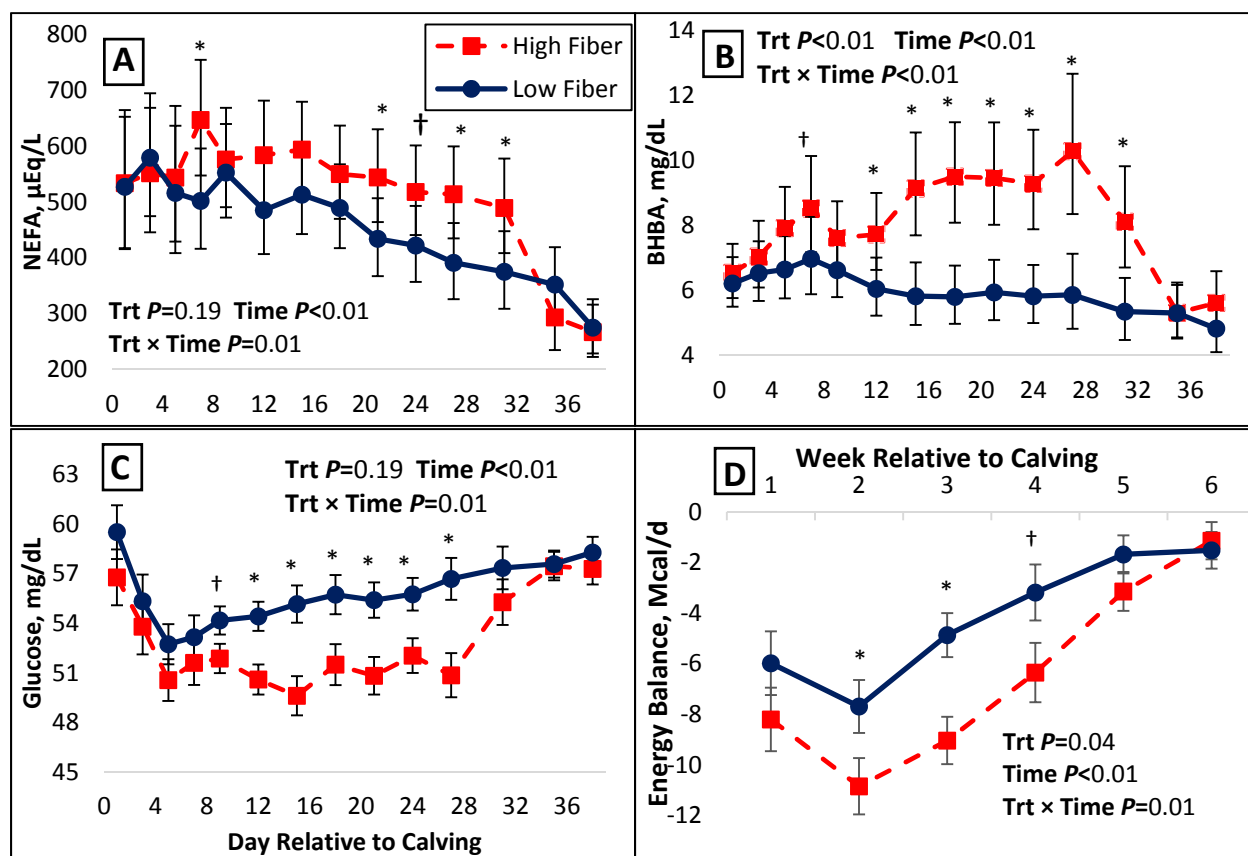


Figure 1. Plasma NEFA (A), BHBA (B), glucose (C), and energy balance (D) for cows fed high fiber and low fiber diets postpartum. Significant differences indicated with an asterisk (*), trends with a cross (†). From LaCount et al., 2017.

The results of these two studies, while not establishing definitive limits or ranges, are critical to begin understanding the role of uNDF₂₄₀ in fresh cow diets. More research in this area is needed, including characterization of all pools of NDF to understand how

we can utilize different pools to affect the performance and metabolism of the transition cow.

SUMMARY AND CONCLUSIONS

In summary, replacement of forage NDF with nonforage fiber sources during the prepartum period generally has increased prepartum DMI, but has translated into minimal differences in postpartum performance or metabolism. In contrast, research generally suggests that increasing forage NDF digestibility during the transition period results in higher prepartum DMI along with higher postpartum DMI, milk yield, and better energetic status. Increased fiber digestibility needs to be accompanied by sufficient uNDF₂₄₀ intake in order to maintain ruminal adaptation to postpartum diets, although more controlled research is needed to refine recommendations for NDF pools for use in diet formulation during the immediate postpartum period.

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